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Experiment: 7

Aim: Implement the embedding wallet (Metamask) and transaction using Solidity.

Theory:

1. Introduction to MetaMask

MetaMask is a widely used cryptocurrency wallet and gateway to blockchain-based applications. It enables users to securely manage digital assets and interact with decentralized platforms directly through a web browser.

MetaMask acts as a bridge between the user's browser and blockchain networks such as Ethereum and other compatible chains.

2. Key Features of MetaMask

- **Multi-Account Management**
  Users can create and manage multiple blockchain accounts within a single wallet.
- **Secure Private Key Storage**
  Private keys are stored locally and encrypted, ensuring enhanced security.
- **Test Network Support**
  Integration with test networks like Sepolia Testnet, Linea, and RSK.
- **DApp Interaction**
  Enables seamless interaction with decentralized applications (DApps).
- **Smart Contract Integration**
  Works efficiently with development tools like Remix IDE for deploying and testing smart contracts.

3. Understanding Testnets

A **Testnet** is a blockchain network designed specifically for testing and development purposes. It replicates the functionality of a real blockchain but uses virtual currency instead of real funds.

Common Testnets

- Sepolia Testnet
- Linea Testnet
- RSK Testnet

4. Advantages of Using Testnets

- **No Financial Risk**
  Transactions use test tokens, eliminating real monetary loss.

- **Safe Testing Environment**
  Ideal for developers to experiment and debug.
- **Pre-Deployment Validation**
  Helps test smart contracts before deploying them on the main network.

5. Steps to Connect MetaMask with Remix IDE

Step 1: Set Up MetaMask

- Install the MetaMask browser extension.
- Create a new wallet or import an existing one using a recovery phrase.

Step 2: Add a Test Network

- Open MetaMask → *Settings* → *Networks* → *Add Network*
- Enter network details (e.g., Sepolia)

Example RPC URL:

https://sepolia.infura.io/v3/<your_project_id>

Step 3: Fund Your Wallet

- Use a faucet to receive free test Ether.
- Example: 0.5 ETH/day (varies by faucet)
- Verify the balance in MetaMask.

Step 4: Connect MetaMask to Remix IDE

- Open Remix IDE
- Navigate to **Deploy & Run Transactions**
- Select **Injected Web3** as the environment
- Approve the connection request in MetaMask

6. Creating and Deploying a Smart Contract

Sample Solidity Contract

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Counter {
  uint public count;

  function increment() public {
     count += 1;
```

```
  }

  function getCount() public view returns (uint) {
    return count;
  }
}
```

Deployment Steps

- Compile the contract in Remix
- Click **Deploy**
- Confirm the transaction in MetaMask

7. Checking Transaction Details

- Use blockchain explorers like:
  - Etherscan (for Sepolia)
  - RSK Explorer
- Track transaction status using the transaction hash

8. Interacting with the Smart Contract

- Use Remix interface to call:
  - increment() → modifies the state (requires transaction)
  - getCount() → reads the value (no gas required)
- Observe transaction confirmations and state updates on the blockchain

<u>Implementation:</u>

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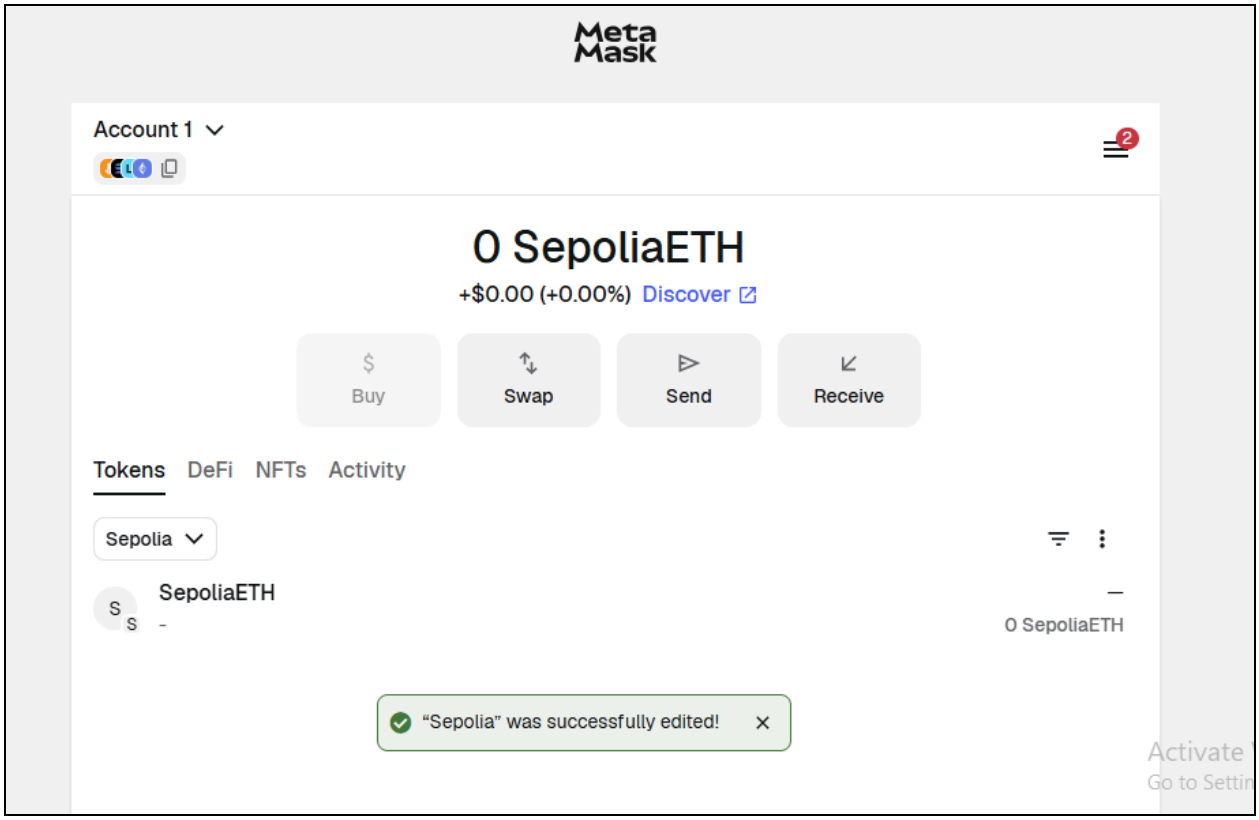
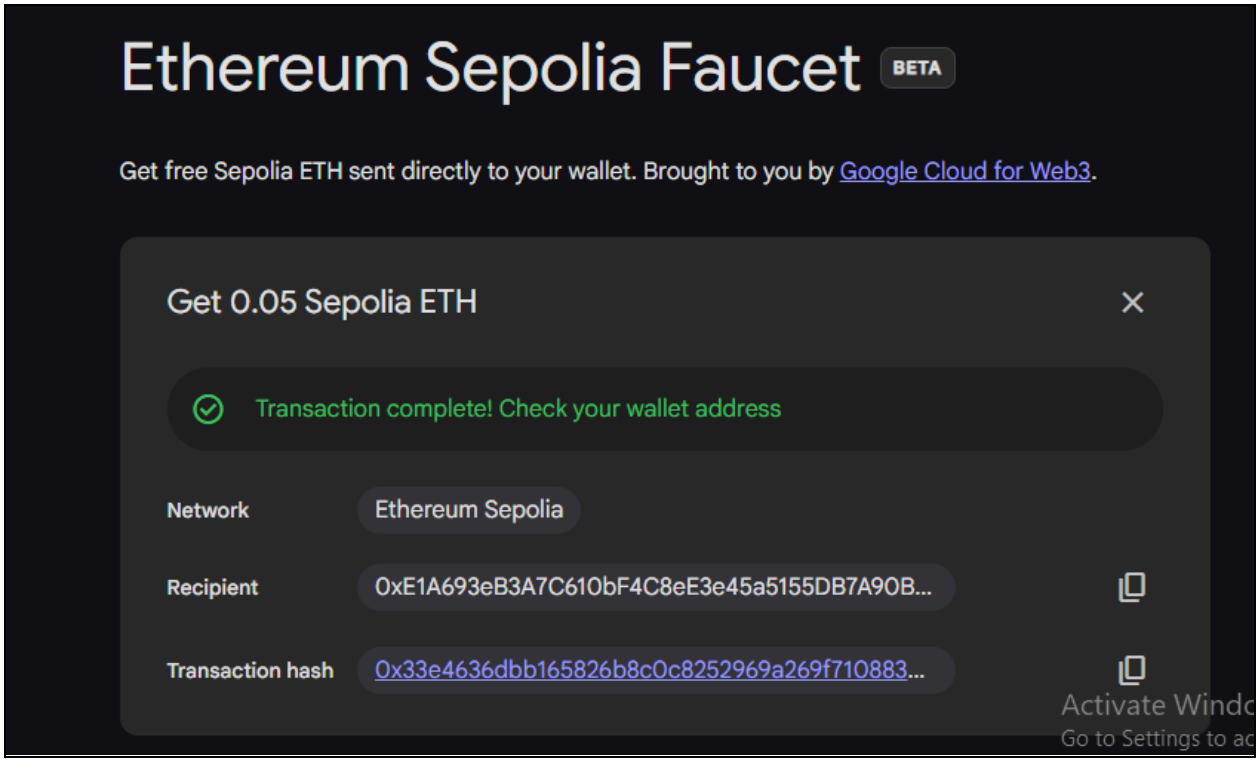


Added the initial balance via google sepolia



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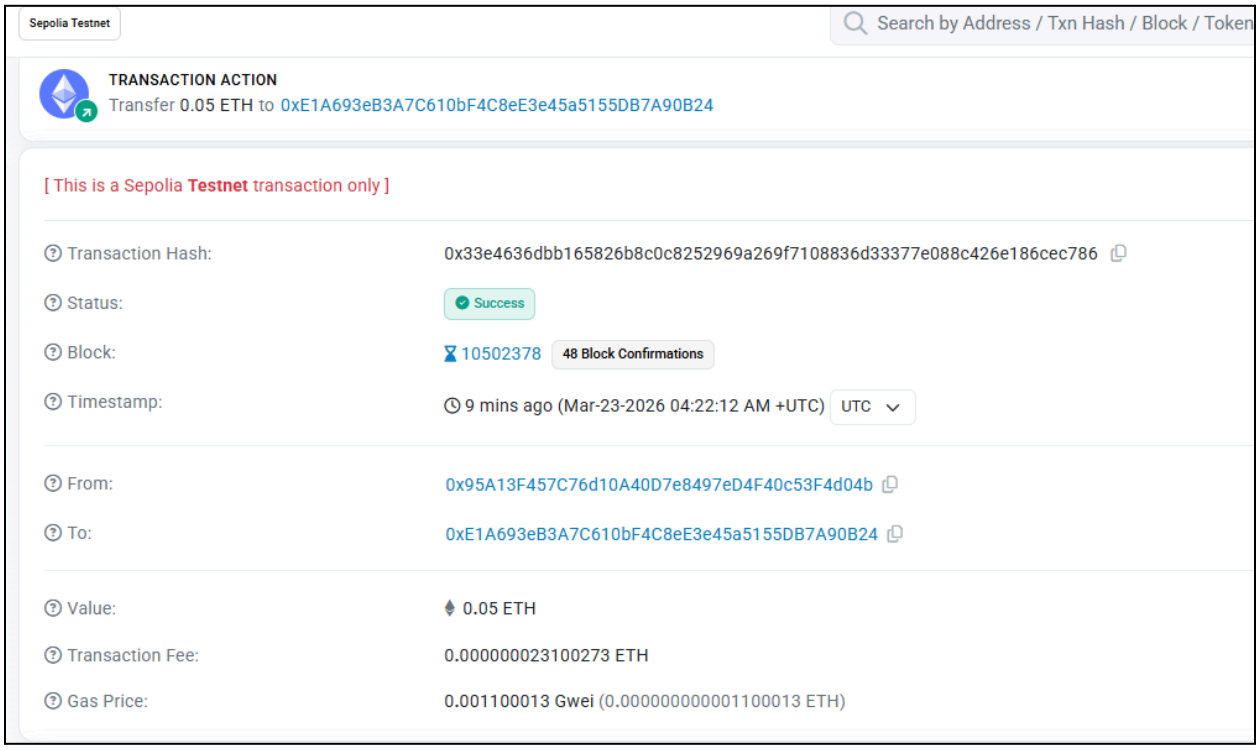
Transaction detail



Linked the remix ide environment to the metamask

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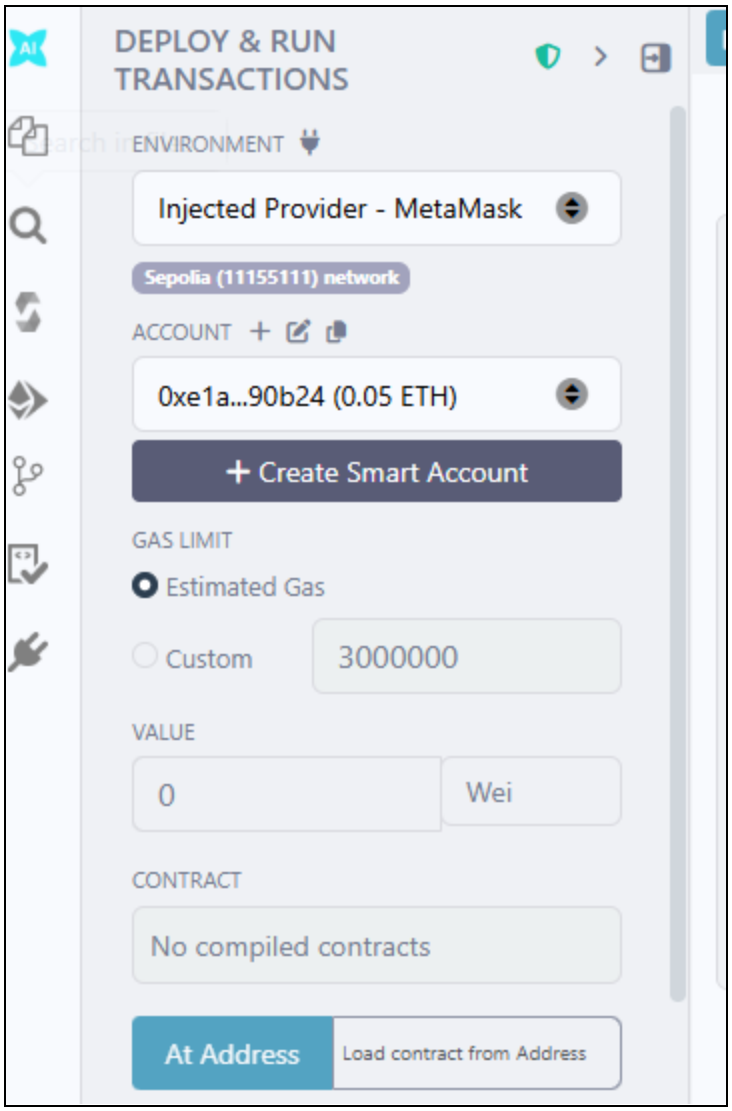

Contract deployed

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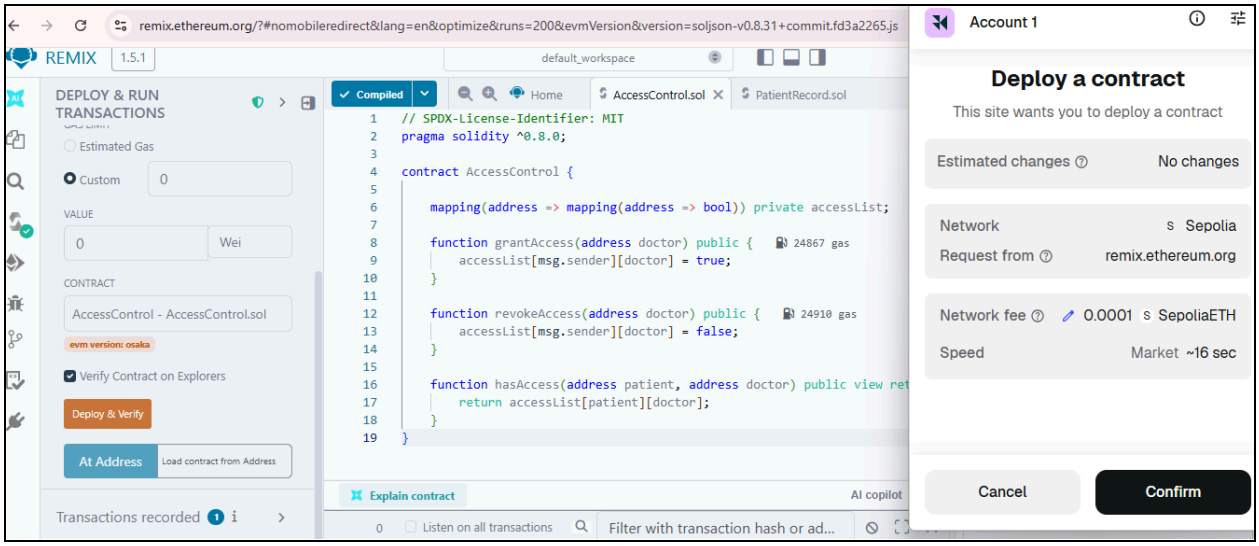

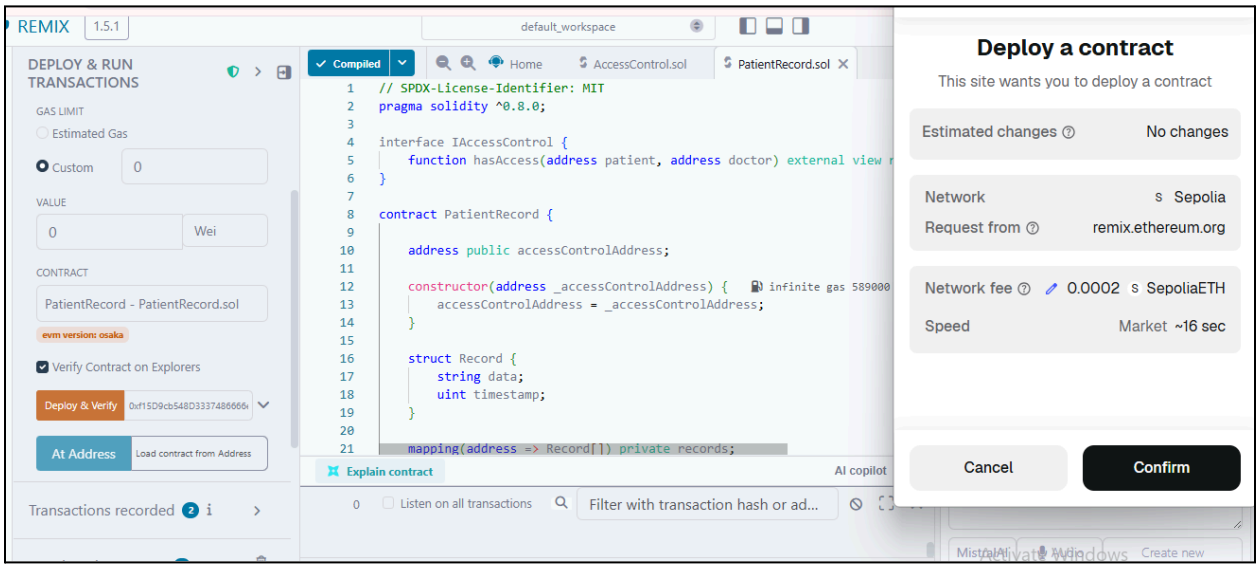

Contract deployed successfully

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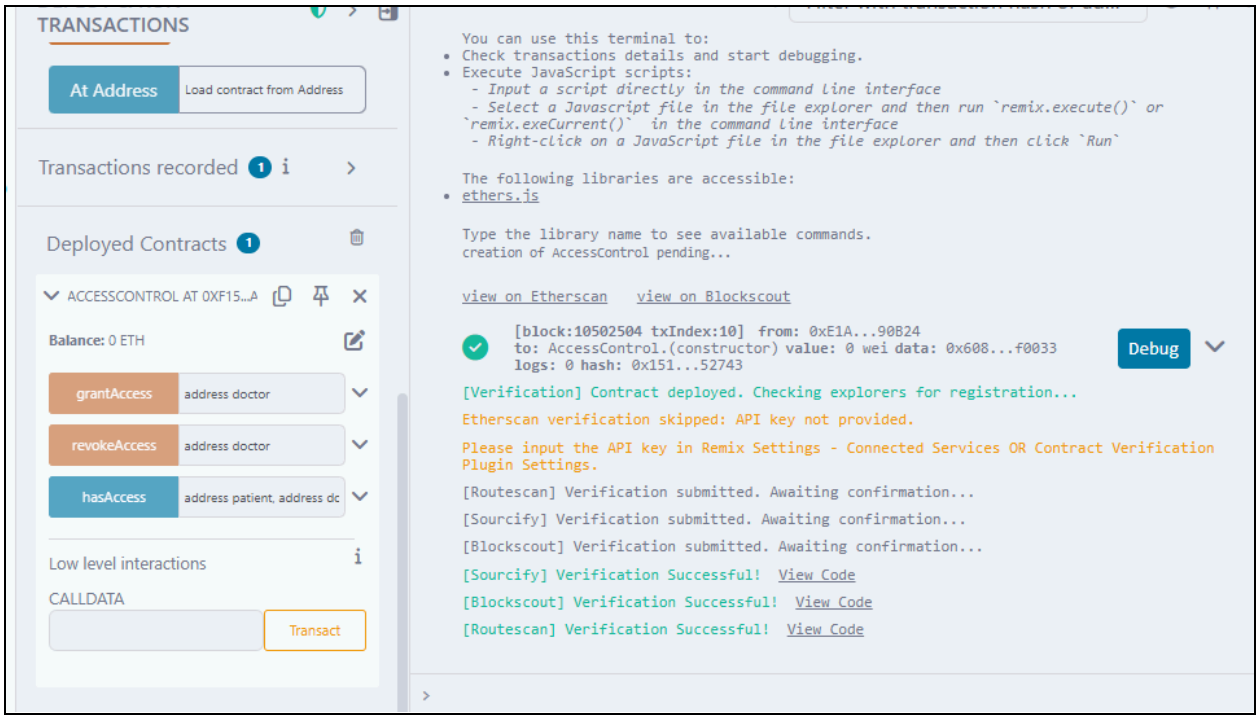
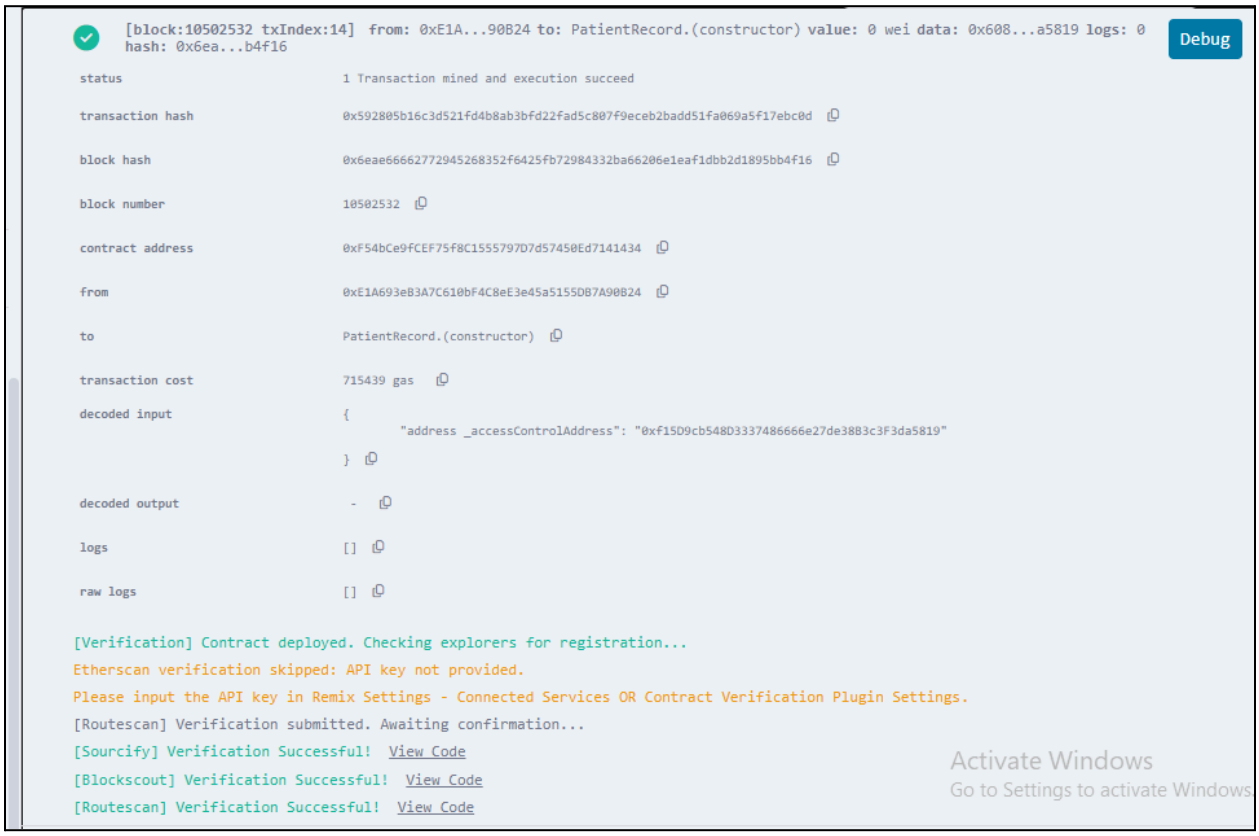


Transaction details in remix ide



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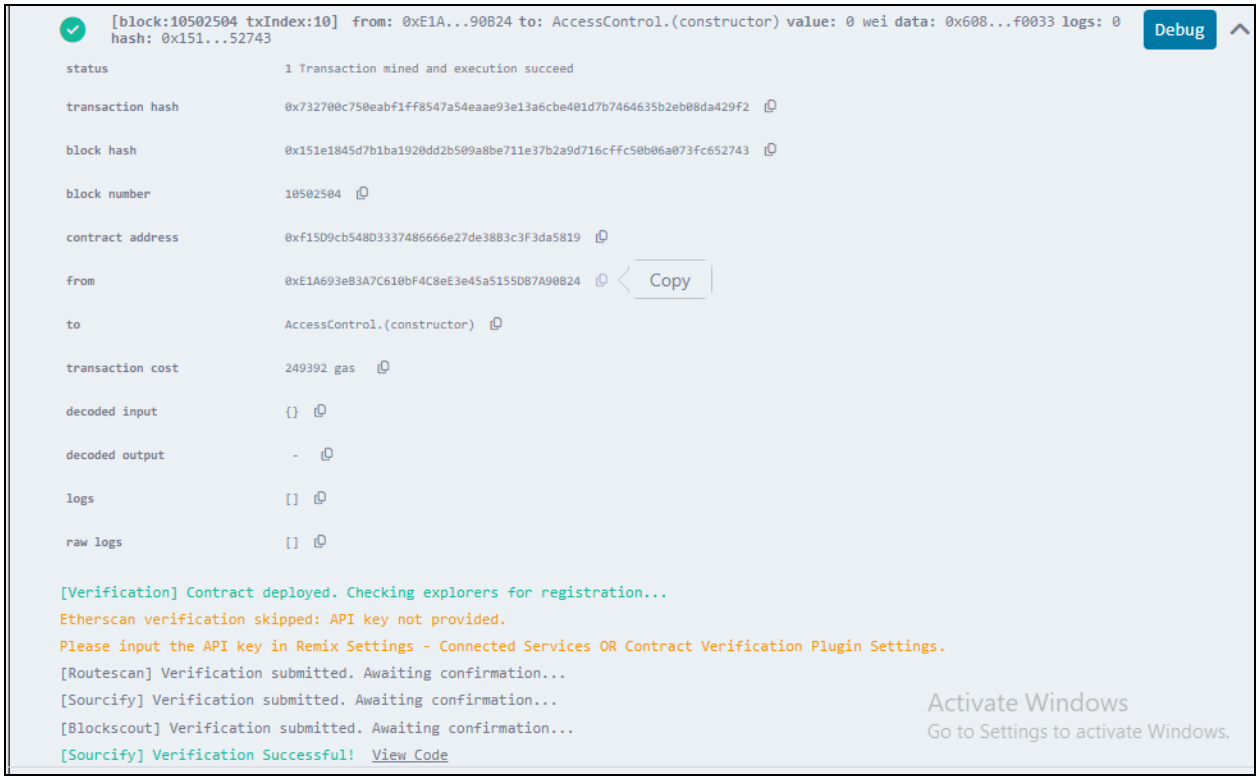
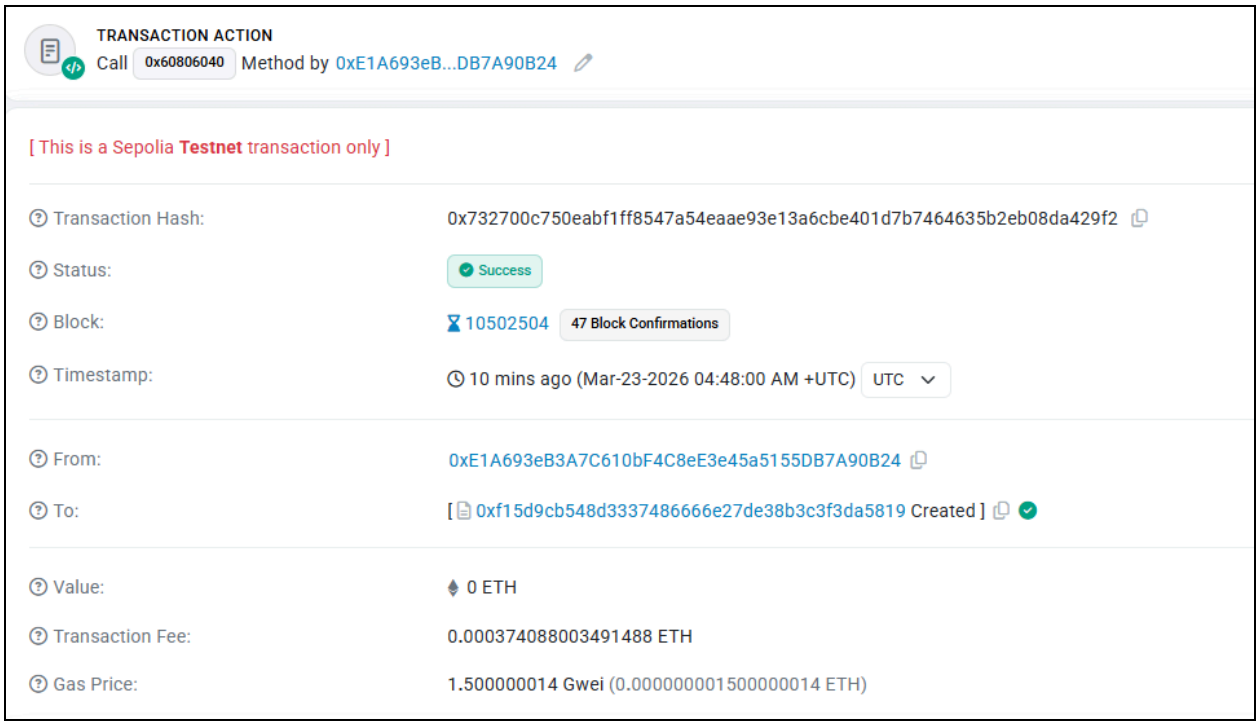


Checking the transactions details



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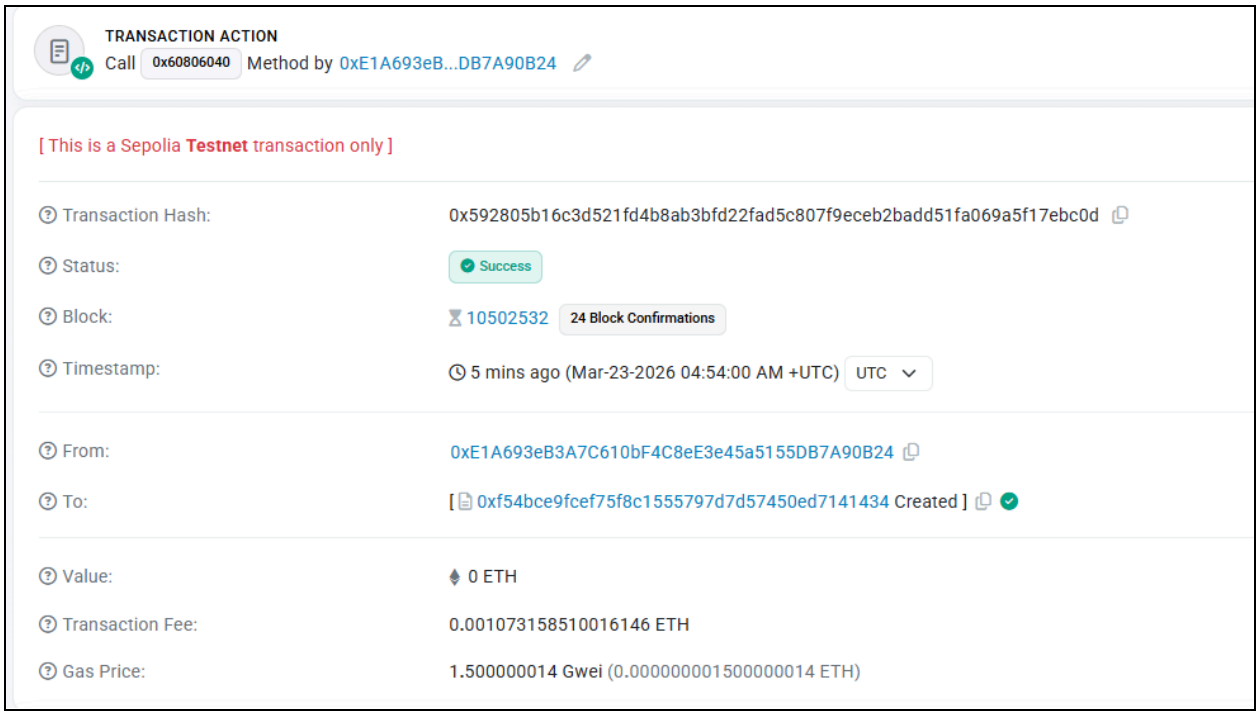
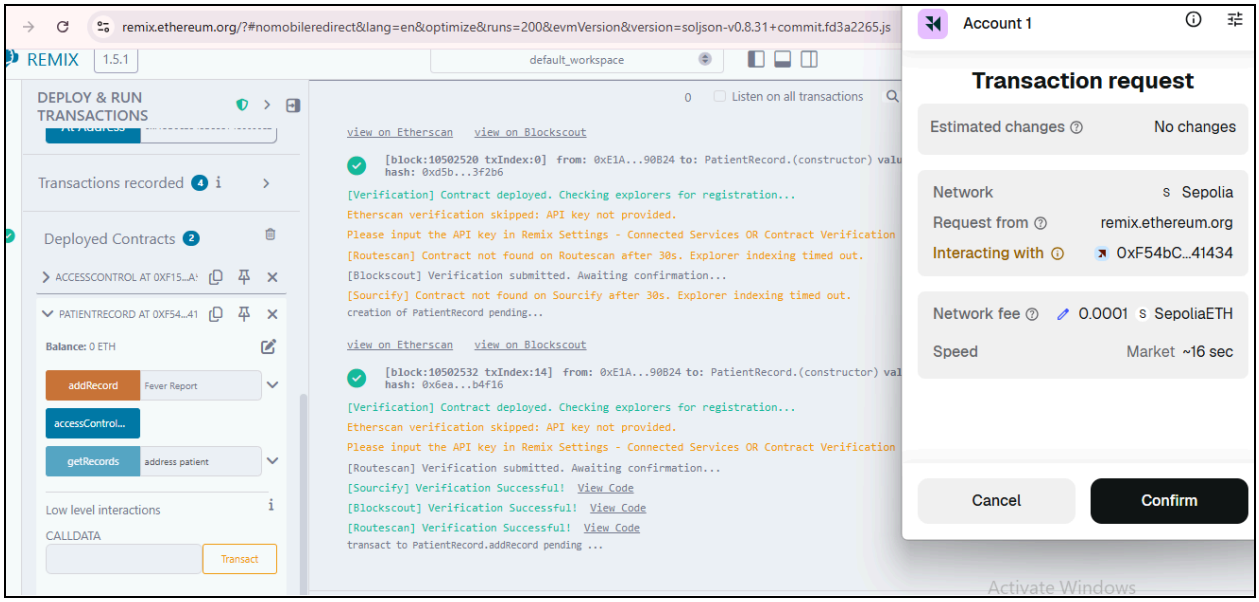


Interaction with smart contracts



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Conclusion:

MetaMask and Remix IDE together provide a powerful environment for blockchain development. By using test networks, developers can safely build, test, and deploy smart contracts without financial risk before moving to the main network.